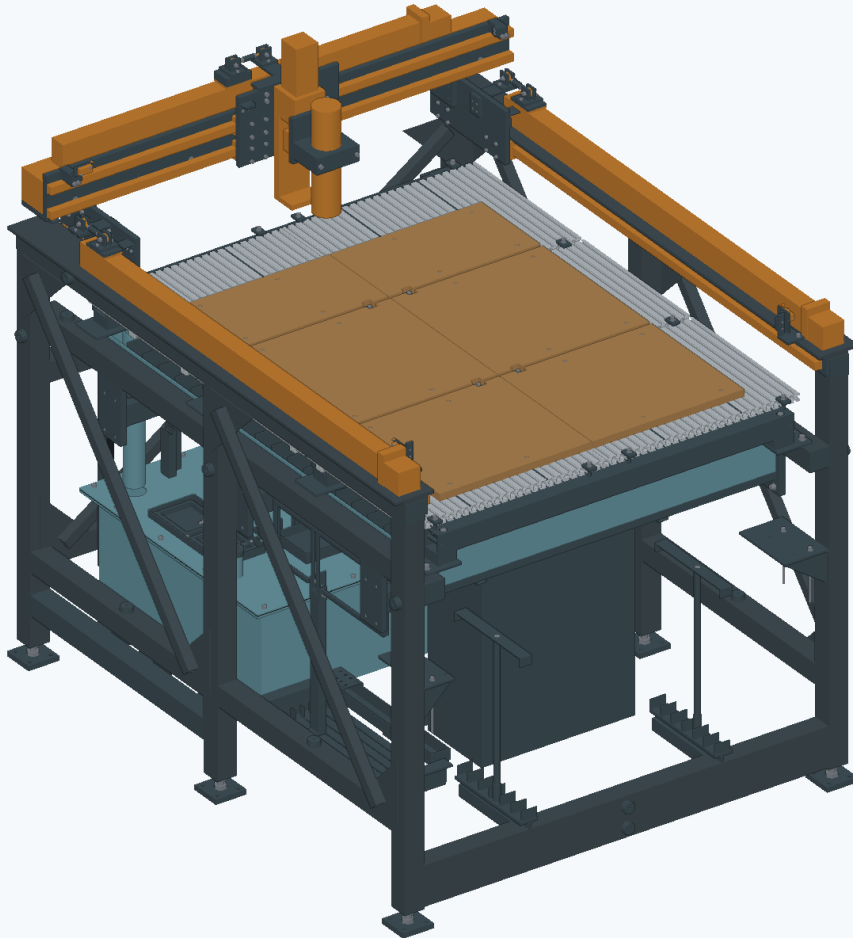


Refining the machine you will build.

Actual Rev I CAD. The Y modules are ordered; the X and Z purchase is planned for Monday. The chassis remains based on 2-inch square tube.



ROUTER BED INSTALLED

Rendered from exported geometry. Amber parts represent purchased-component envelopes; several mating dimensions still require measurement.

800 X / 1000 Y / 100 Z

Nominal travel. Both selected HMS40 X/Y listings specify 10 mm lead: 10 mm movement per screw revolution. This is not screw diameter.

2 x 2-inch steel chassis

Current model wall: 0.120 inch (3.048 mm). Scrap lengths, wall and condition are still unknown. Sand can add stationary mass; it does not supply structural stiffness.

What this revision changes

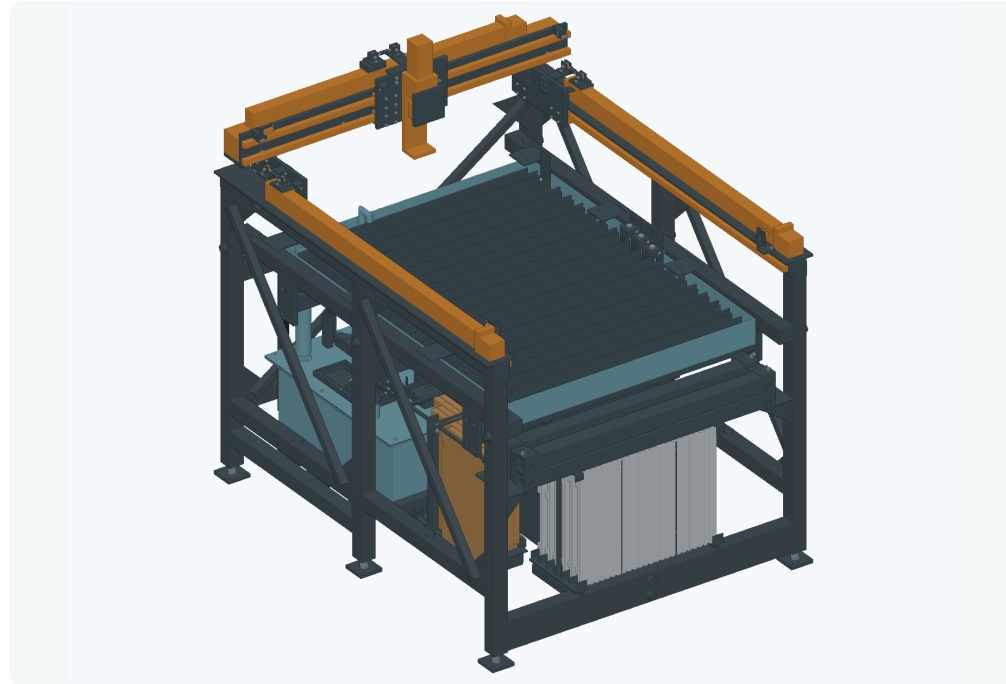
Stronger storage retainers, a guided float/breakaway head, supported plumbing, service access and actual control circuits replace several unresolved details.

Design review, not finished machinery.

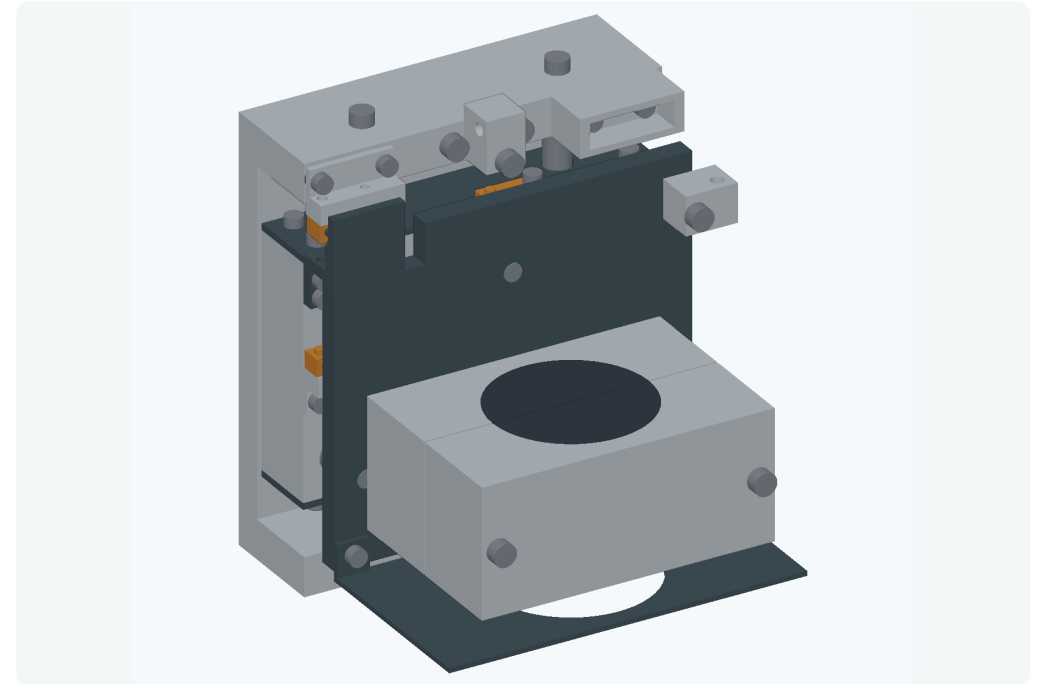
The head mechanism is modeled. Actual torch and module interfaces, whole-machine rigidity and electrical qualification remain open.

Conversion stays inside the frame.

Manual panel-and-beam conversion preserves the floor footprint. It retains 52 primary release fasteners plus restraint and tool-exchange steps.



STORED BED / ISOMETRIC



FLOAT / BREAKAWAY DETAIL / UNBORED INSERT

Scope: the stored view contains the bed and removed router tools. The head hardware view has no invented torch. Clear the front storage racks before servicing the controls cabinet.

01 / Captive hardware

Twelve nut strips retain the spoilboard hardware. Thin slot lips cannot accept ordinary torque-table preload: the actual extrusion needs the specified coupon test before release.

02 / Defined storage

The two storage rods grow from 10 to 14 mm with 40 mm receiver engagement, thicker seats and a revised clearance path. The 100 N single-rod bump screen now has a 2.25 yield reserve.

03 / Supported tool exchange

One rear panel stays clamped as a work surface. Park the router through the open front bay, then split and transfer the head. Restore the front beam before storing the last panel. Follow the detailed sequence.

Dimensions, materials and interfaces.

The steel stock lengths remain unchanged. Service openings and fabricated hardware are modeled; unknown supplier interfaces remain explicitly guarded.

BED, STOCK AND WATER

Bare bed	1003 x 1211 overall; larger than nominal cutting travel
Six removable panels	500 x 397 each; separate reachable spoilboards
Extrusion stock	10 bars at 1220 = five two-packs; 3 x 397 per bar
2 x 2 tube blanks	30 blanks / 29,472 mm / 96.69 ft net
Tube lengths and quantities	4 x 1450; 7 x 1048.4; 3 x 1016.4; 6 x 949.2; 4 x 924; 6 x 648.8
Nominal wall	3.048 mm; do not silently substitute thinner scrap
Water storage	Vented reservoir; gravity drain and electric refill
Refill discharge	Air gap above the pan; verify at least 25 mm as built

Net tube length excludes saw loss and other profiles, plate and sheet. Five full 20-ft lengths have a documented nest; random scrap must be nested from its actual usable lengths. No new material price is asserted.

Supported drain and basket service

A reducer, valve allocation and removable flanged tail replace generic blocks. The tail parks internally, allowing the basket to lift. Exact threaded make-up remains a receipt fit.

Pressure and suction paths

Supported Parker hose routes use 80 mm bends and a separate strainer carrier. Enlarged hose sweeps check 2 mm radial allowance. Pump orientation and tray bores now match the routing.

Rigidity has a stated target

Known bed members account for about 0.135 mm at 100 N against a proposed 0.20 mm tool-to-work target. Actual motion, joint and tool compliance cannot be assigned zero.

One control baseline

Kraken V1.1 retains onboard drivers. Terminal wiring, timers, run isolation and restart prevention are defined. The C41S speed converter uses a 200 Hz commissioning setting.

Evidence and the remaining work.

The checks cover explicit geometry, routes and circuit behavior. Physical supplier fit, cutting rigidity and control commissioning remain separate release conditions.

CHECKED ARTIFACTS

Router state	1669 components; 0 unresolved nominal overlaps
Stored state	1669 components; 0 unresolved nominal overlaps
Head hardware state	1669 components; 0 unresolved nominal overlaps
STEP readback	All three assemblies reimported; matching solid counts and volumes
Source identity	Build manifest records exact source hashes and no changes during generation
Check scope	18 router/head poses, named handling/service paths and 450 bounded circuit checks; see the linked evidence
Limits	Member/joint screens are conditional; full rigidity, physical interfaces and electrical operation are not qualified

Current package: output/release-review/RevI-CAD
Completion evidence: output/design-finish-2026-09-26
Rebuild: build_revi.py in the legacy-named source directory.

Public project: github.com/DeadlySIn777/plasma-router-stand

Y arrival and X/Z order

Measure base slots/nuts, carriage hole pattern, height and stroke datums. The Z listing alone does not resolve its output mounting or prove the optional brake motor will fit.

Actual plasma assembly

The mechanism now has defined guides, stops, locators, magnets and switch cams. VIV ARC CUT-50 version, torch bore, start interface, lead routing and force/stop tests remain required.

Rigid structure with the actual stock

Confirm remaining scrap wall and usable lengths. Qualify the complete frame, bed, gantry, mounts, joints and feet against explicit loads and a tool-to-work deflection target.

Operation and procurement

Complete physical panel layout, stop/power/brake integration, measured hose ends and live interlock tests. Revised quantities supersede historical budgets; no current delivered total is asserted.